

Module 3: Cell division, development and disease control

3.1 Cell division and development

This section provides learners with knowledge and understanding of biological development on three levels: cell division by mitosis (where genetic information is copied and passed to daughter cells) cell differentiation, cell division by meiosis and fetal development, and the evolutionary development of species.

Scientific research has highlighted the role of apoptosis during the life cycle of a multicellular organism. The importance of both mitosis and apoptosis is considered in relation to fetal growth and development, as well as an appreciation that these processes are affected by environmental factors. Knowledge of recent advancements in stem cell technology and its potential applications further enhance learners' understanding of these fundamental cellular processes.

Meiosis is also studied as a process that promotes genetic biodiversity and variation. The potential for change and the development of new species by the action of natural selection on genetic variation is seen as an essential part of the process of evolution. Learners will also gain an understanding that the variety of life, both past and present, is extensive, but the biochemical basis of life is similar for all living things as well as of the link between natural selection and the adaptations of living organisms to their environments.

Learners are expected to apply knowledge, understanding and other skills gained in this section to new situations and/or to solve related problems.

3.1.1 The developing cell: cell division and cell differentiation

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) the cell cycle	To include the processes taking place during interphase (G_1 , S and G_2), mitosis and cytokinesis, leading to genetically identical daughter cells.
(b) (i) the changes that take place in the nuclei and cells of animals and plants during mitosis	To include the changes in the nuclear envelope and the behaviour of the centrioles, spindle fibres, centromere, chromatids and chromosomes, and the formation of the cell plate in plant cells.
(ii) the microscopic appearance of cells undergoing mitosis	To include the examination and drawing of stained sections or squashes of plant tissue and the identification of the stages observed. PAG1 HSW4, HSW5, HSW6
(c) the principal stages and features of apoptosis	To include cell shrinkage, nuclear condensation (pyknosis), blebs, nuclear fragmentation (karyorrhexis), the roles of phosphatidylserine and macrophages.
(d) the importance of apoptosis and mitosis in growth and repair	To include examples of the roles of apoptosis in cell deletion and mitosis in cell addition.

(e) **(i)** the differentiation of stem cells into specialised cells

To include an appreciation of the differences between totipotent, pluripotent and multipotent stem cells, and the differentiation of bone marrow stem cells into specialised blood cells.

(ii) current applications and uses of stem cells.

To include the use of bone marrow stem cells.